R CODE LAB EXPERIMENT

SET 14 – SURVEY RESULTS

```
respondent <- 1:3
q1 <- c("A","B","C")
q2 <- c("B","A","D")
q3 <- c("C","A","B")

survey <- rbind(table(q1),table(q2),table(q3))

barplot(survey,
        beside=FALSE,
        col=rainbow(4),
        legend=rownames(survey),
        main="Survey Responses",
        xlab="Questions",
        ylab="Count")

install.packages("fmsb")
library(fmsb)

radar <- data.frame(
A=c(3,1,2),
B=c(3,2,1),
C=c(3,1,1),
D=c(3,0,1)
)
```

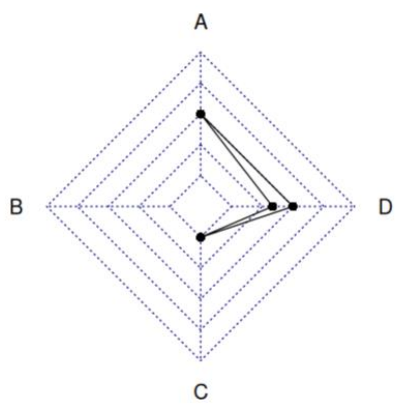
R CODE LAB EXPERIMENT

```
radarchart(radar)

survey_table <- data.frame(
Respondent=respondent,
Question1=q1,
Question2=q2,
Question3=q3
)

print(survey_table)
```

```
> print(survey_table)
  Respondent Question1 Question2 Question3
1          1         A         B         C
2          2         B         A         A
3          3         C         D         B
>
```



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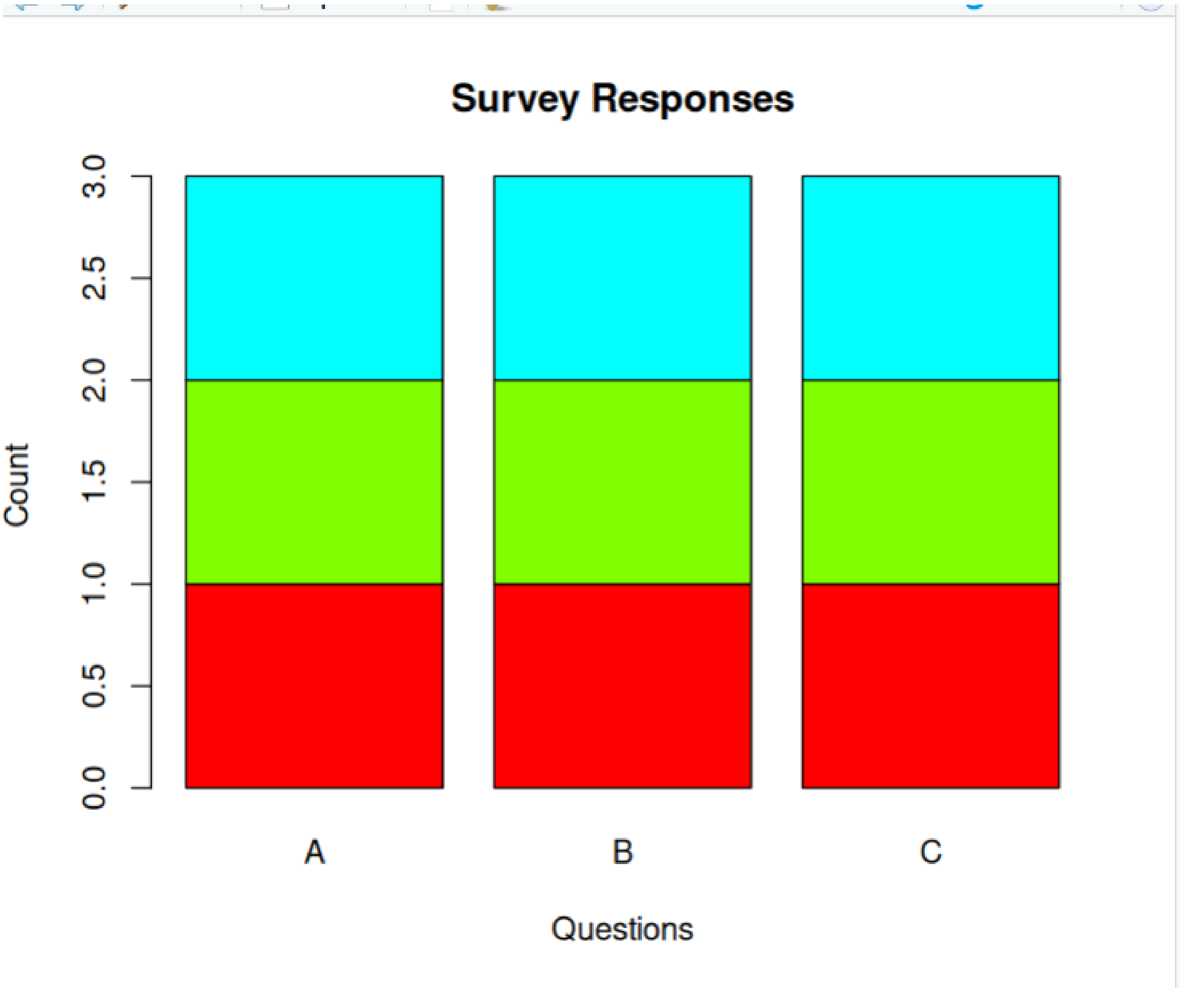


SET 15 – PRODUCT SALES ANALYSIS

```
product <- c("Product A","Product B","Product C")

jan <- c(2000,1500,1200)

feb <- c(2200,1800,1400)

mar <- c(2400,1600,1100)


sales <- data.frame(

January=jan,

February=feb,

March=mar
```

R CODE LAB EXPERIMENT

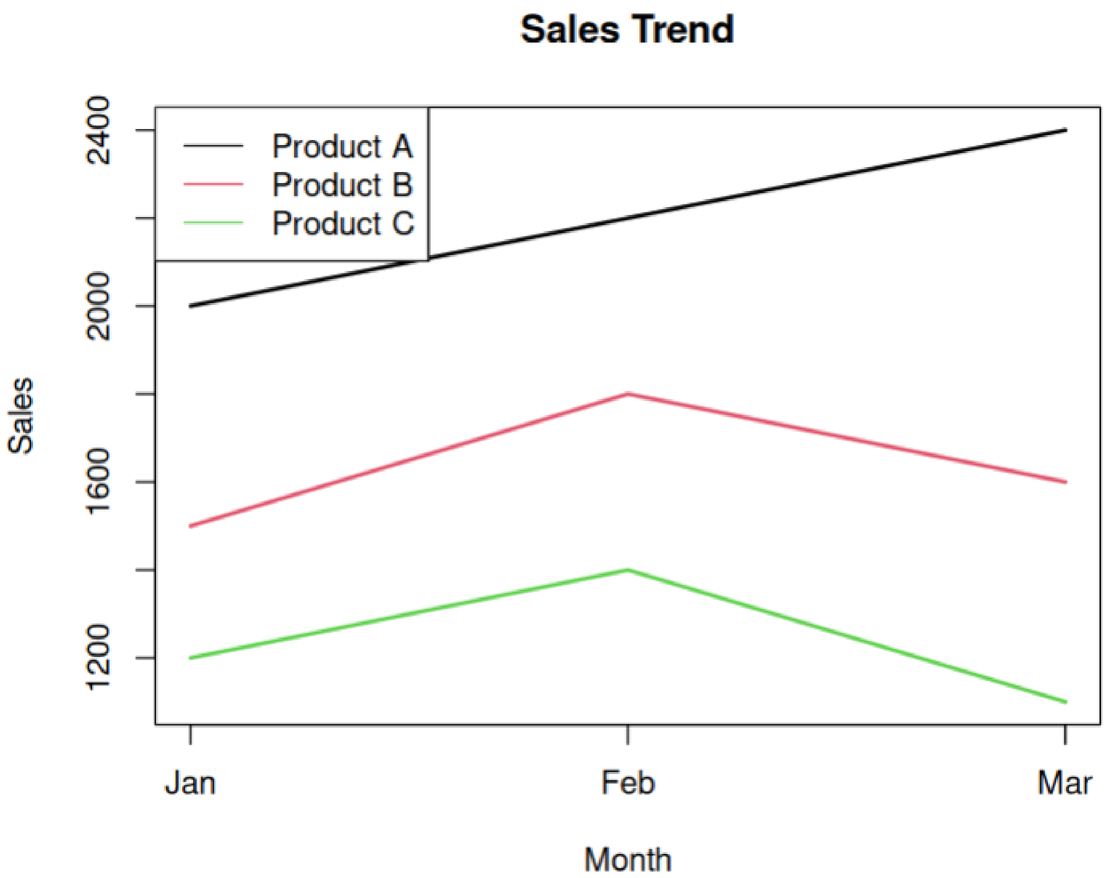
```
)

rownames(sales) <- product

barplot(as.matrix(sales),
        beside=TRUE,
        col=c("red","green","blue"),
        legend=colnames(sales),
        main="Quarterly Product Sales",
        xlab="Products",
        ylab="Sales")

matplot(t(sales),
        type="l",
        lty=1,
        lwd=2,
        col=1:3,
        xaxt="n",
        xlab="Month",
        ylab="Sales",
        main="Sales Trend")

axis(1,at=1:3,labels=c("Jan","Feb","Mar"))

legend("topleft",
legend=product,
col=1:3,
```

R CODE LAB EXPERIMENT

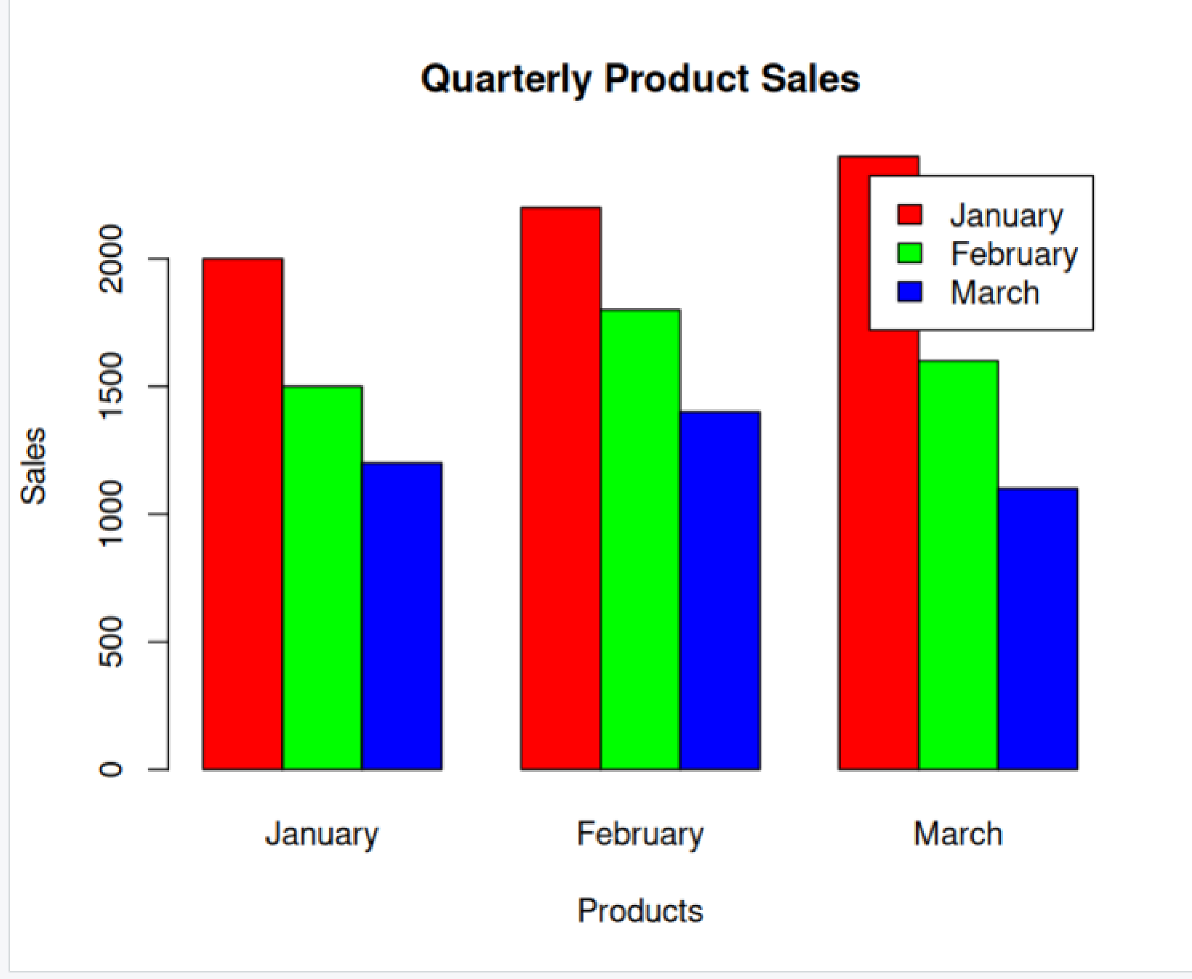
lty=1)

print(sales)

```
          January February March
Product A    2000     2200  2400
Product B    1500     1800  1600
Product C    1200     1400  1100
>
```



R CODE LAB EXPERIMENT



SET 16 – CUSTOMER DEMOGRAPHICS ANALYSIS

customer <- 1:3

age <- c(28,35,42)

gender <- c("Female","Male","Female")

income <- c(50000,60000,75000)

barplot(age,

names.arg=customer,

col="skyblue",

R CODE LAB EXPERIMENT

main="Customer Age Distribution",

xlab="Customer ID",

ylab="Age")

pie(table(gender),

col=c("pink","lightblue"),

main="Gender Distribution")

demo <- data.frame(

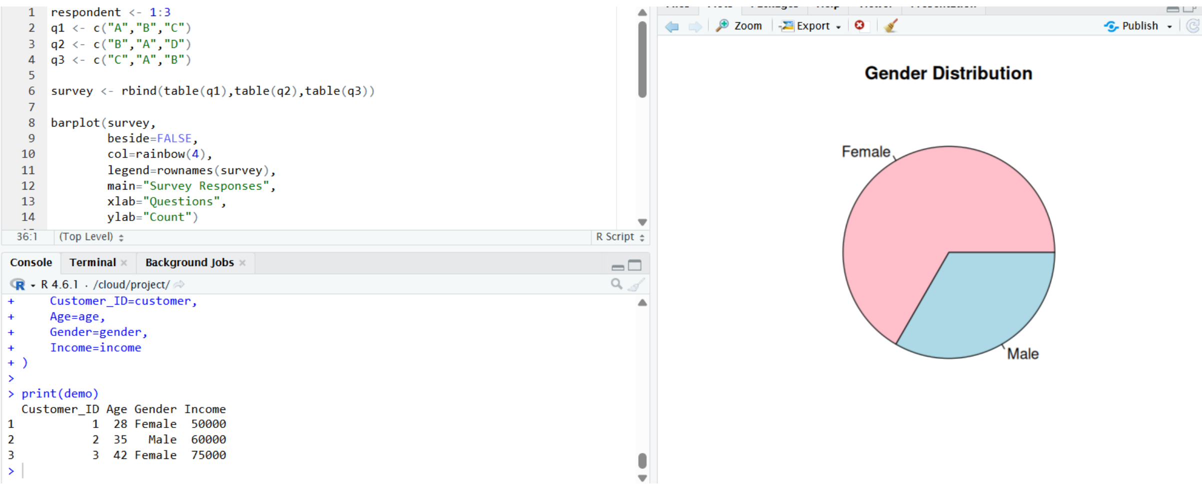
Customer_ID=customer,

Age=age,

Gender=gender,

Income=income

)



R CODE LAB EXPERIMENT

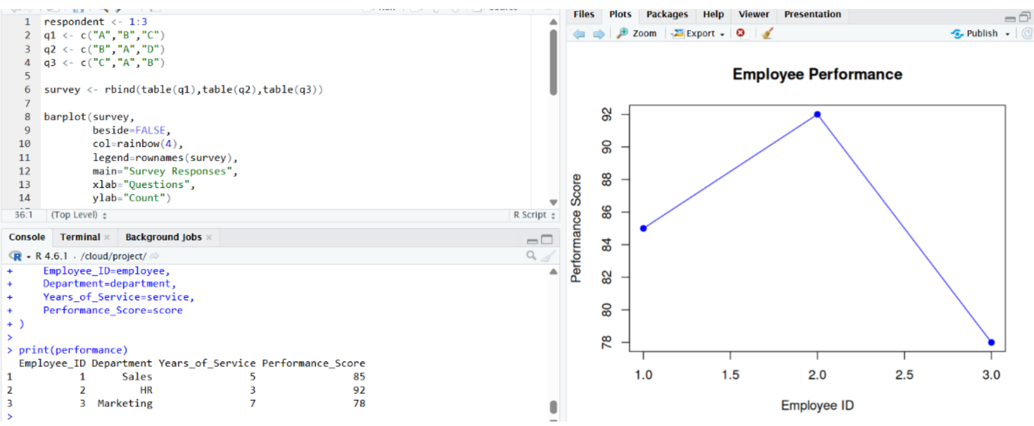
print(demo)

SET 17 – EMPLOYEE PERFORMANCE ANALYSIS

```
employee <- 1:3
department <- c("Sales","HR","Marketing")
service <- c(5,3,7)
score <- c(85,92,78)
```

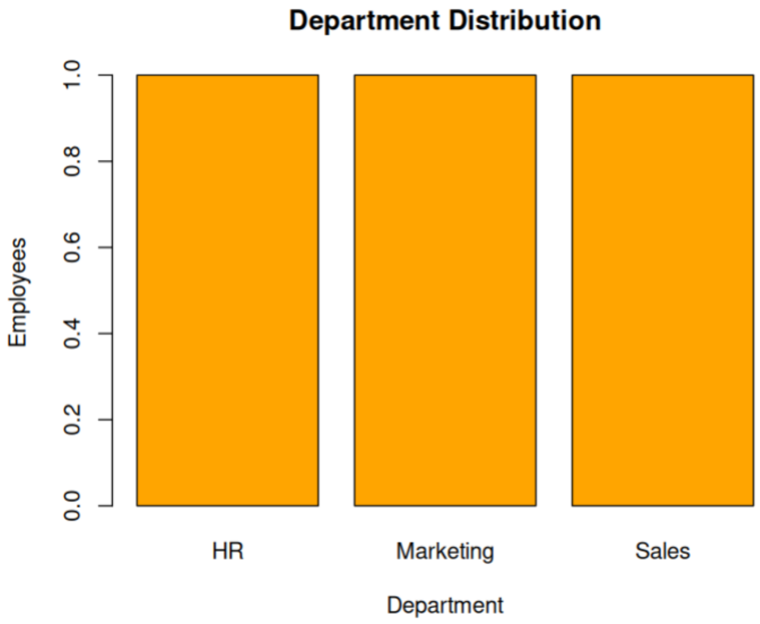
R CODE LAB EXPERIMENT

```
plot(employee,
score,
type="o",
pch=19,
col="blue",
main="Employee Performance",
xlab="Employee ID",
ylab="Performance Score")

barplot(table(department),
col="orange",
main="Department Distribution",
xlab="Department",
ylab="Employees")

performance <- data.frame(
Employee_ID=employee,
Department=department,
Years_of_Service=service,
Performance_Score=score
```




```
)
```

print(performance)



SET 18 – PRODUCT INVENTORY MANAGEMENT

product <- c("Product A","Product B","Product C")

quantity <- c(250,175,300)

price <- c(20,15,18)

barplot(quantity,

names.arg=product,

col="green",

main="Inventory Quantity",

xlab="Products",

ylab="Quantity")

category1 <- c(120,90,150)

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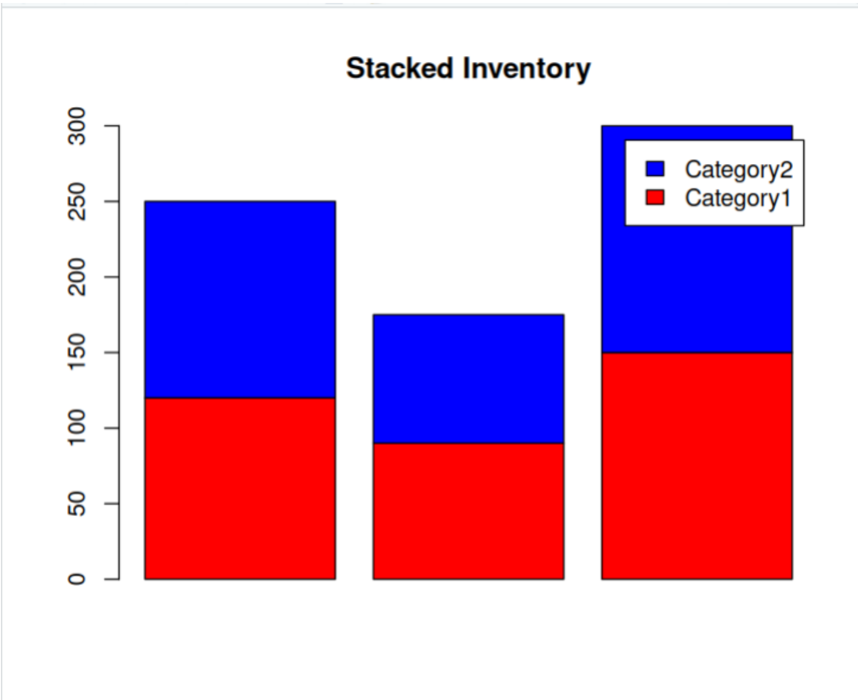
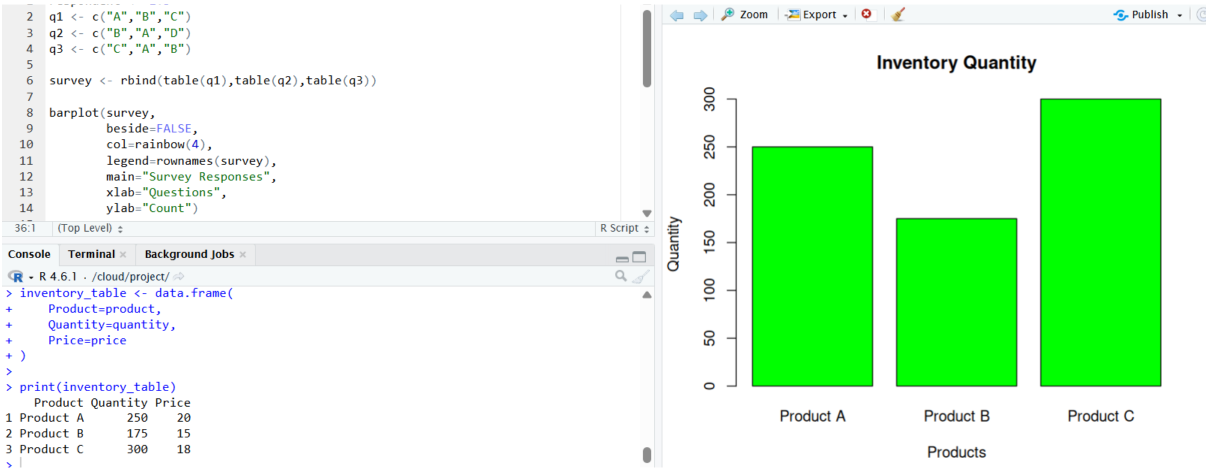
```
category2 <- quantity-category1

inventory <- rbind(category1,category2)

barplot(inventory,
col=c("red","blue"),
legend=c("Category1","Category2"),
main="Stacked Inventory")

inventory_table <- data.frame(
Product=product,
Quantity=quantity,
Price=price
)

print(inventory_table)
```



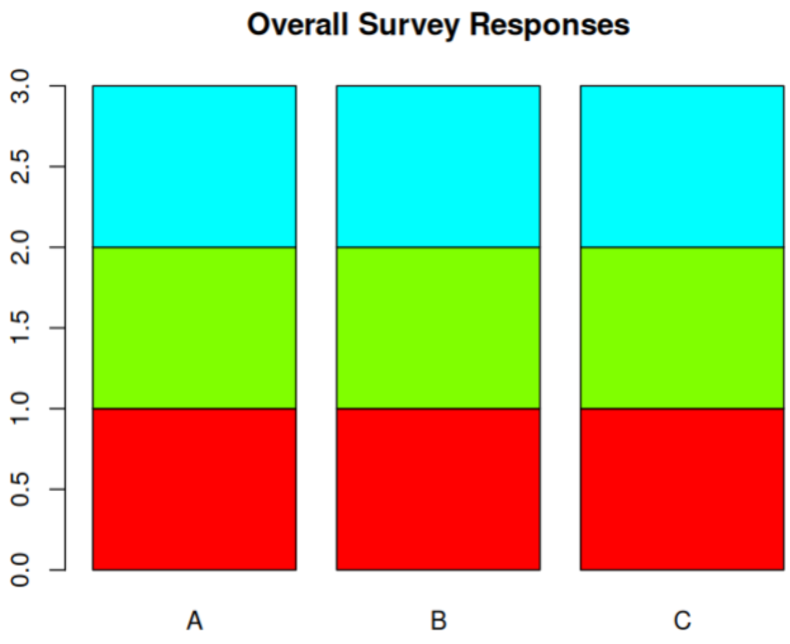
R CODE LAB EXPERIMENT



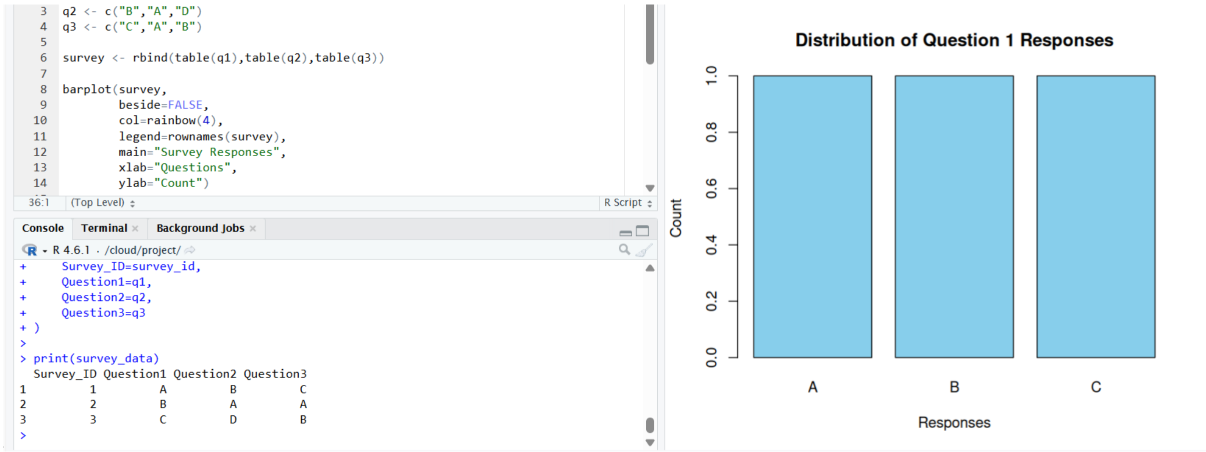
SET 19 – SURVEY RESPONSES ANALYSIS

```
survey_id <- 1:3
q1 <- c("A","B","C")
q2 <- c("B","A","D")
q3 <- c("C","A","B")

q1_table <- table(q1)

barplot(q1_table,
        beside=TRUE,
        col="skyblue",
        main="Distribution of Question 1 Responses",
        xlab="Responses",
        ylab="Count")

response_table <- rbind(table(q1),table(q2),table(q3))

barplot(response_table,
```

```
    beside=FALSE,
    col=rainbow(4),
    legend=rownames(response_table),
    main="Overall Survey Responses")

survey_data <- data.frame(
Survey_ID=survey_id,
Question1=q1,
Question2=q2,
Question3=q3
)

print(survey_data)
```



R CODE LAB EXPERIMENT



SET 20 – STOCK ANALYSIS

```
date <- as.Date(c("2023-01-01","2023-01-02","2023-01-03"))


stockA <- c(100,105,110)
stockB <- c(150,152,148)
stockC <- c(120,118,122)


plot(date,
    stockA,
    type="o",
    col="red",
    ylim=c(90,160),
    xlab="Date",
    ylab="Stock Price",
    main="Stock Price Analysis")


lines(date,stockB,type="o",col="blue")
lines(date,stockC,type="o",col="green")


legend("topleft",
```

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```
legend=c("Stock A","Stock B","Stock C"),
col=c("red","blue","green"),
lty=1,
pch=1)

change <- c(0,diff(stockA)/head(stockA,-1)*100)

barplot(change,
names.arg=date,
col="orange",
main="Daily Percentage Change - Stock A",
xlab="Date",
ylab="% Change")

stock_data <- data.frame(
Date=date,
Stock_A=stockA,
Stock_B=stockB,
Stock_C=stockC
)
```

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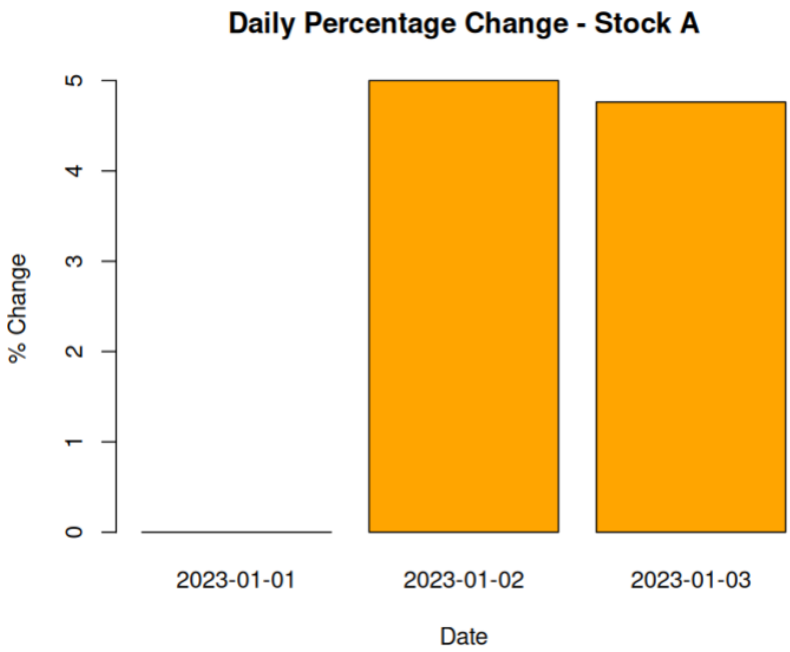
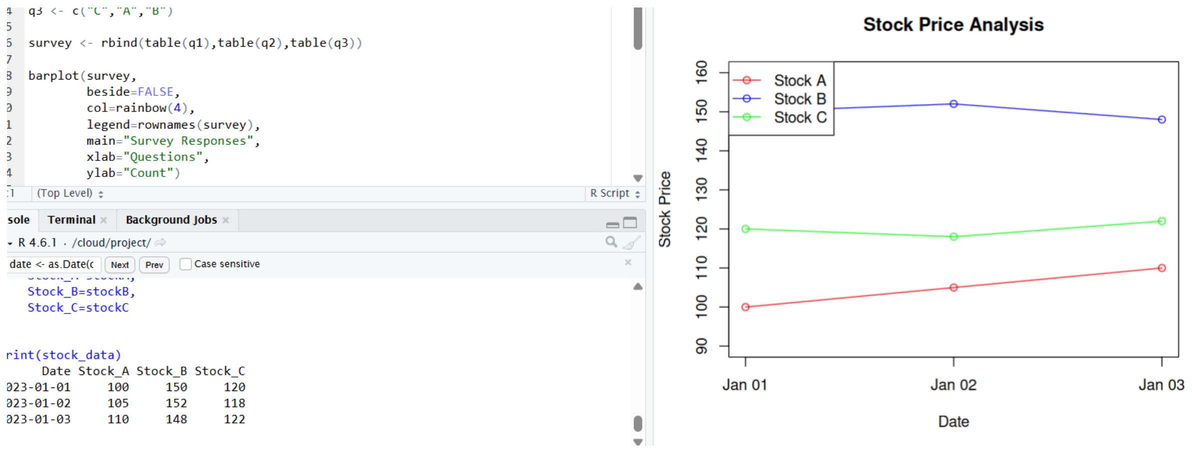
print(stock_data)





SET 21 – ENERGY CONSUMPTION DATA

sector <-
c("Residential","Commercial","Industrial","Residential","Commercial","Industr
ial")

R CODE LAB EXPERIMENT

```
region <- c("North","South","West","East","North","South")
month <- c("Jan","Jan","Feb","Feb","Mar","Mar")
temperature <- c(15,24,20,18,28,30)
units <- c(320,540,880,350,610,920)
cost <- c(2100,3600,5900,2300,4100,6200)
renewable <- c(22,18,12,25,20,15)
peak <- c(4,6,8,5,7,9)

hist(units,
probability=TRUE,
col="skyblue",
main="Units Consumed",
xlab="Units Consumed")

lines(density(units),col="red",lwd=2)

symbols(temperature,
units,
circles=peak,
inches=0.3,
bg=rgb(0,0,1,0.5),
xlab="Temperature",
ylab="Units Consumed",
main="Temperature vs Units")

avg <- aggregate(renewable,
by=list(sector),
```

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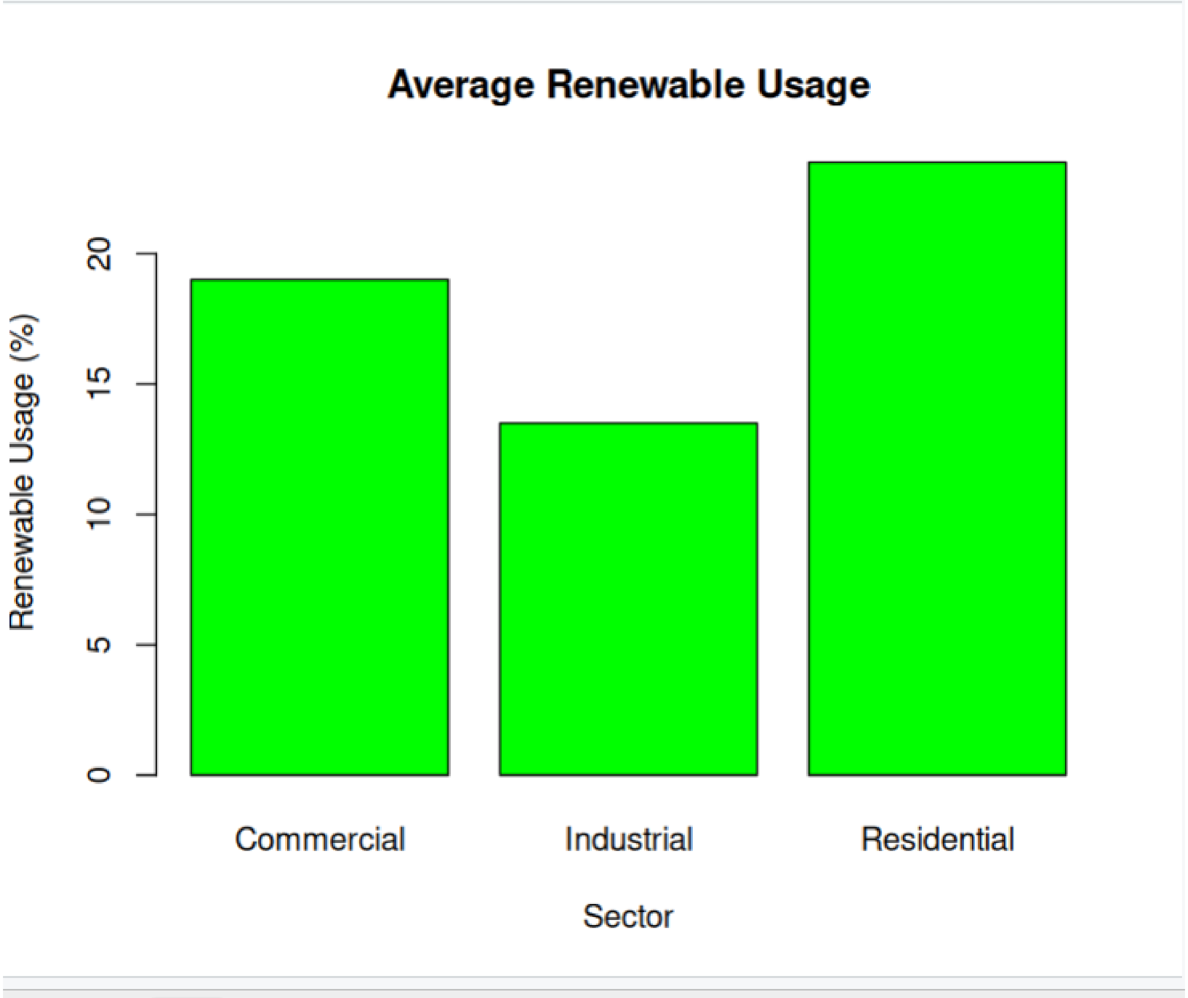
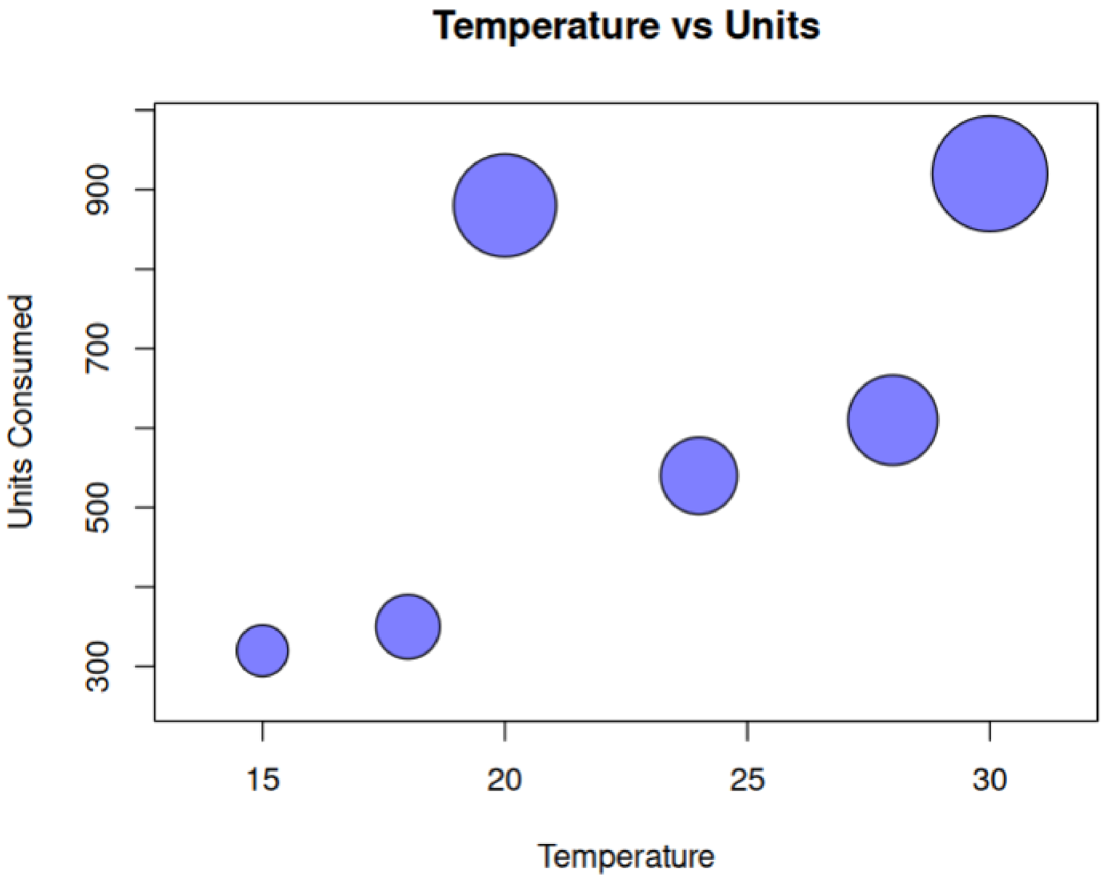
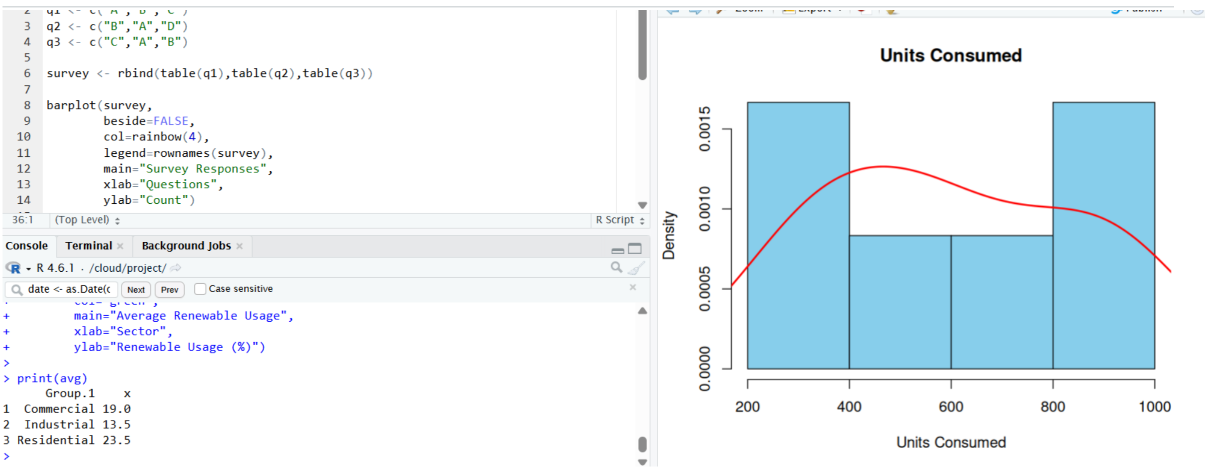
```
FUN=mean)

barplot(avg$x,
names.arg=avg$Group.1,
col="green",
main="Average Renewable Usage",
xlab="Sector",
ylab="Renewable Usage (%)")

print(avg)
```



R CODE LAB EXPERIMENT





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SET 22 – TIME SERIES ANALYSIS

```
month <- c("January","February","March","April","May")
sales <- c(15000,18000,22000,20000,23000)

ts_sales <- ts(sales,start=1,frequency=12)

plot(ts_sales,
type="o",
col="blue",
xlab="Month",
ylab="Sales",
main="Monthly Sales Trend")

advertising <- c(2000,2500,3000,2800,3200)

plot(advertising,
sales,
pch=19,
col="red",
xlab="Advertising Budget",
ylab="Sales",
main="Advertising Budget vs Sales")

acf(ts_sales,
main="Autocorrelation Plot")
```

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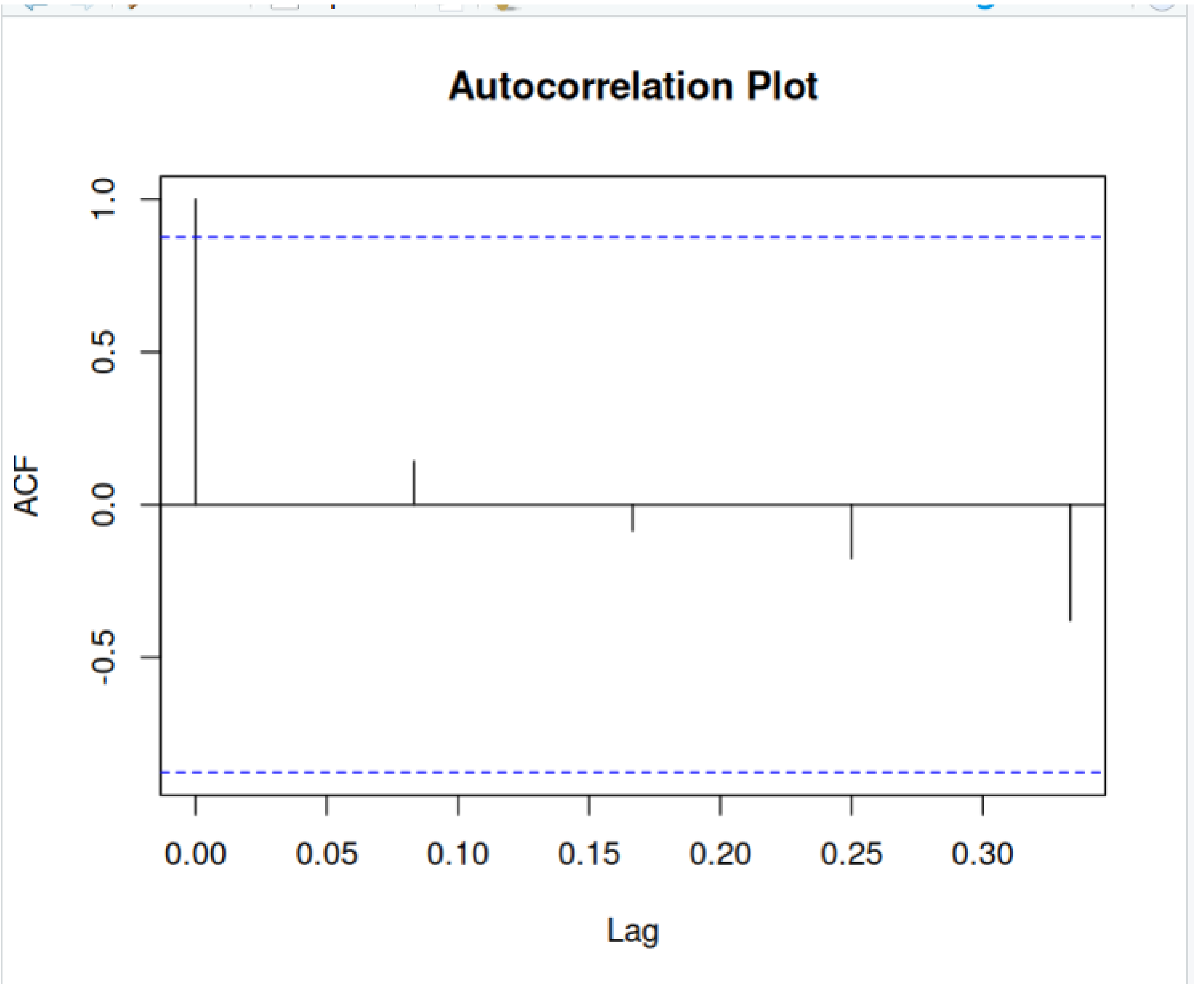
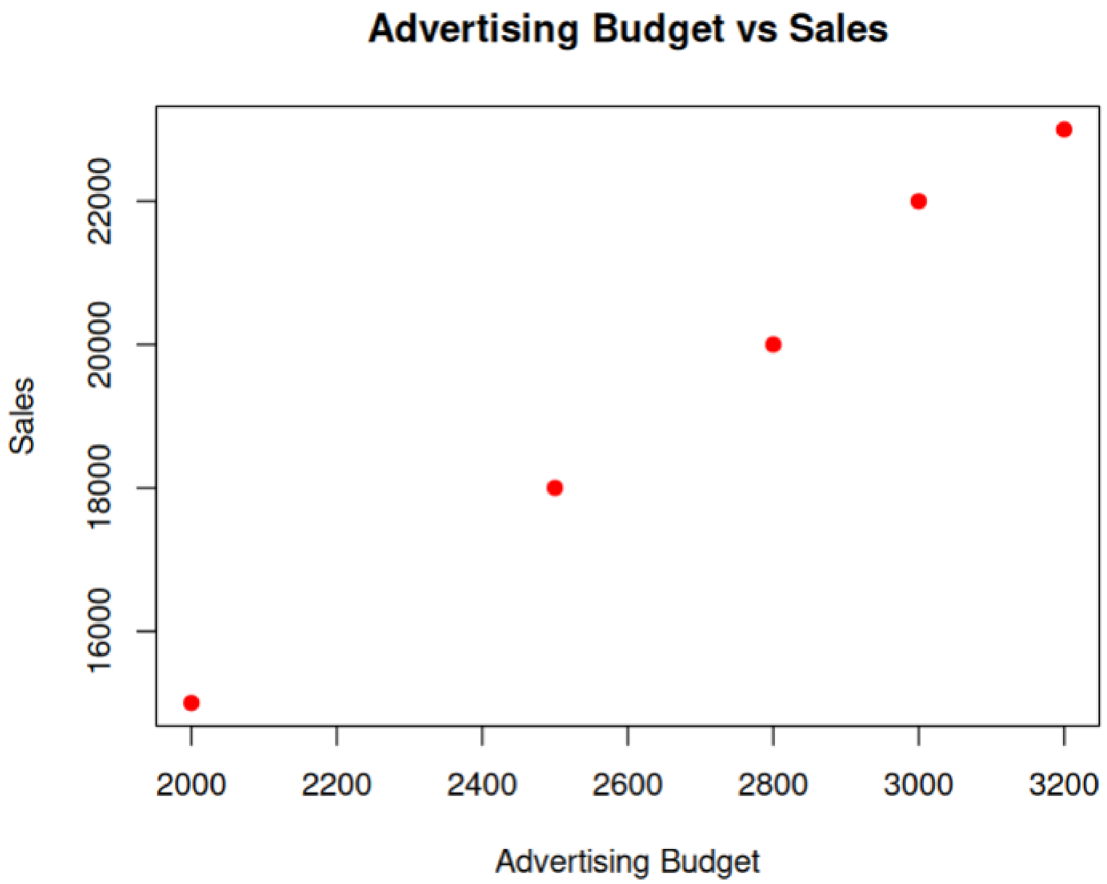
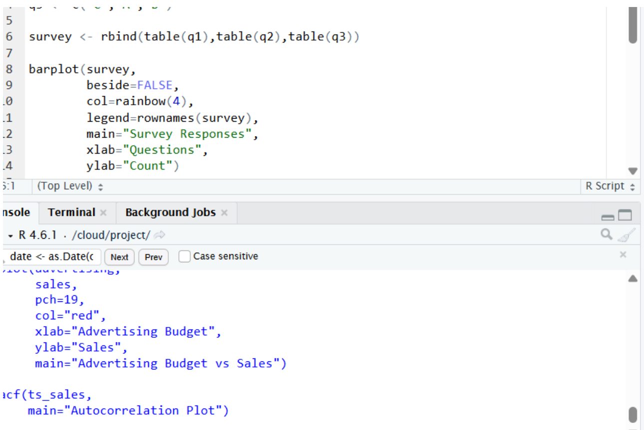
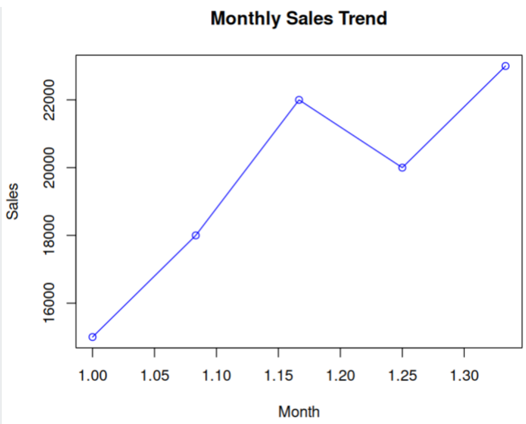


R CODE LAB EXPERIMENT







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SET 23 – EMPLOYEE PERFORMANCE ANALYSIS

```
employee <- 1:5
department <- c("Sales","HR","Marketing","Sales","HR")
service <- c(5,3,7,4,2)
score <- c(85,92,78,90,76)

barplot(table(department),
col="orange",
main="Department Distribution",
xlab="Department",
ylab="Employees")

plot(employee,
score,
type="o",
pch=19,
col="blue",
xlab="Employee ID",
ylab="Performance Score",
main="Employee Performance")

performance <- data.frame(
Employee_ID=employee,
Department=department,
Years_of_Service=service,
```

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Performance_Score=score

)

print(performance)